

Housing units, household choices, and equilibrium prices

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The question. Why does the model offer houses of size 2, 4, 6 rather than 0.2, 0.4, 0.6, or 20, 40, 60? The numerical scale is a choice of units. It should not affect consumption, saving, tenure, fertility, or physical price responses. Our equations admit that equivalence, provided quantities, prices, and utility coefficients are converted together. Changing the grid alone would change the economic model.

What one housing unit means. The empirical quantity anchor is the American Housing Survey count of rooms, rather than bedrooms. The model uses that room scale for housing quantities and assumes services proportional to quantity, with a separate owner service premium. A four-room house is $H = 4$ in room units. Define $h = H/10$ if one housing unit instead represents ten rooms; the same house is then $h = 0.4$. Its physical size has not changed. The empirical room target must also be divided by ten when expressed in these new units.

Consumption and affordability. Income and consumption remain in the same monetary units. If rent per room is r , rent per ten-room unit is $10r$. The household therefore pays exactly the same amount:

$$rH = (10r)(H/10).$$

The same identity applies to purchase value, down payments, collateral, resale proceeds, depreciation, and property taxes. All these costs use price times housing quantity, with unchanged financing and tax rates. Consequently the same consumption and saving choices remain feasible. As an illustration, not a calibration estimate:

	Rooms	Ten-room units
Same dwelling's quantity	4	0.4
Rent per housing unit	10	100
Total rent paid	40	40
Income available	100	100
Consumption after rent and a child expense of 10	50	50

Fertility and utility. Preserving the budget is necessary but not sufficient: the value of consumption and housing must retain its comparison with the value of children. For a fixed consumption weight α , write material utility as

$$u^{\text{mat}}(c, H) = \frac{[c^\alpha H^{1-\alpha}]^{1-\sigma}}{1-\sigma}.$$

Here c is consumption and σ is curvature of utility over the consumption–housing composite. One transparent conversion replaces H by $10h$ in this expression. Material utility then stays exactly the same, and child utility needs no alteration.

Owners use housing directly; renters use a closed-form allocation in which the same unit conversion enters through rent. An equivalent conversion is therefore to keep its functional form in h and multiply *all utility-valued coefficients* by the common factor $K = (0.1)^{(1-\alpha)(1-\sigma)}$. With $\alpha = 0.733$ and $\sigma = 2$, $K = 1.849$. This includes child benefits, the fixed birth cost, bequest strength, and taste-shock scales. Total values then satisfy $V' = KV$, while the ratios of value differences to taste-shock scales remain unchanged. Birth-attempt and tenure probabilities, and optimal consumption and saving, are consequently unchanged. Re-estimation is not needed for this conversion.

Why the derivative can change without changing behavior. A derivative is measured per numerical housing unit. An upgrade from four to six rooms is $4 \rightarrow 6$ in room units and $0.4 \rightarrow 0.6$ in ten-room units. The numerical derivative and the numerical increment change together. The finite utility comparison for the same physical upgrade remains unchanged. For example, holding family state and tenure fixed and ignoring a housing floor, the consumption sacrifice at material indifference for $4 \rightarrow 6$ is 13.730% under the baseline weight 0.733. It is also 13.730% for $0.4 \rightarrow 0.6$ after a consistent conversion.

Housing supply and price curvature. Supply is expressed in the same housing units as demand. Let Q denote total housing, H_0 its supply scale, r unit rent, $\bar{r}$ reference unit rent, and ξ supply elasticity:

$$Q = H_0(r/\bar{r})^\xi, \quad r = \bar{r}(Q/H_0)^{1/\xi}.$$

Moving to ten-room units divides Q and H_0 by ten and multiplies r and $\bar{r}$ by ten. The equation remains satisfied at exactly the same physical allocation. Rescaling H_0 is part of this conversion; it does not require making H_0 an endogenous equilibrium variable.

Inverse supply is convex when $\xi < 1$. Nevertheless its proportional price response is constant:

$$\frac{d \log r}{d \log Q} = \frac{1}{\xi}.$$

At the specified $\xi = 0.630$, a 5% quantity expansion along the supply curve requires an 8.052% rent increase, in either set of units. The numerical slope changes with the coordinates; the implied change in rent per actual room does not. This calculation concerns the supply curve, not the full equilibrium response to a preference or fertility shock, which also depends on demand and the population closure.

Application to the two implemented specifications. The adopted September 28 reference changes the housing weight with children and compensates material utility using a fixed reference rent. That reference rent must also be multiplied by ten. The compensation then restores the same common factor K across family states; scaling the grid and child benefit alone is insufficient. In the later floor experiment, the housing weight is common and the physical floor must be divided by ten along with the grid. Its population closure sets population equal to total supplied housing divided by demand per household. Both quantities convert together, leaving population and birth renewal unchanged. These are separate specifications, not interchangeable calibration results.

What the audit establishes. A read-only review traced the implemented budget, material utility, child benefit, bequest, choice-probability, aggregation, and supply equations. Scalar arithmetic on transcribed formulas reproduced the conversion identities at illustrative points, including family compensation and the physical floor; it did not evaluate the renter closed-form allocation or compare model outputs. Historical parameter conversions remain unchecked. Source hashes cover the floor experiment, not the adopted reference. No model was imported, recalibrated, or solved. A full numerical equivalence certificate remains outstanding because the code also contains fixed feasibility thresholds and numerical regularizers.

Conclusion. Using rooms is a coherent normalization: the numerical label 2 rather than 0.2 does not itself select a different region of economic curvature. What must be retained is the meaning of the house and every trade-off it enters. This result does not validate proportional services per room, continuous fractional rental quantities, financial affordability, or the imposed demand and supply elasticities. Those are substantive assumptions and empirical checks; a change of units neither fixes nor invalidates them. The source map and arithmetic record are in the accompanying housing-unit audit.

From model prices to dollars: a worked example

The two units. Housing is measured in rooms; money is measured relative to the model’s baseline annual gross earnings scale, which is normalized to one. Let M be the number of 2007 dollars corresponding to one model monetary unit. Let p be the purchase price of one room and r its rent for one four-year model period. The model links them by $r = up$, where u is the period user-cost coefficient. Neither r nor u determines the earnings-to-dollar conversion.

Where the dollar scale comes from. The diagnostic uses PSID 2007 gross labor earnings of the reference person and spouse combined, not total household income. For reference persons aged 18–65 with nonnegative earnings and positive survey weights, it takes weighted means within twelve four-year age groups and then averages those twelve means equally, matching the model income profile’s age-cell normalization. The result is \$77,963.06 in 2022 purchasing power. Using the retained CPI-U annual indices converts this to 2007 dollars:

$$M \simeq 77,963.06 \times \frac{207.342}{292.655} \simeq \$55,235.75 \quad \text{per model monetary unit.}$$

This conversion is obtained from earnings, independently of the housing prices being checked; it was not chosen to make those prices match the AHS.

Purchase prices and rental payments. Purchase value is a stock, so multiply p by M . Rent is a payment over four years, so multiply r by M and divide by 48 months. Using the full saved reference values gives:

$$\begin{aligned} \text{Purchase value per room} &= 0.7898695 \dots \times M = \$43,629.03, \\ \text{Monthly rent per room} &= 0.1423972 \dots \times M/48 = \$163.86. \end{aligned}$$

The same conversion applied to the floor experiment gives the last column:

Cost per room	AHS 2007	Sept. 28 reference	Oct. 1 floor
Monthly rent (\$)	168.22	163.86	149.20
Purchase value (\$)	43,495.49	43,629.03	39,723.80

The data column uses weighted total annual contract rent divided by weighted renter rooms and by twelve; owner value uses weighted total reported dwelling value divided by weighted owner rooms. These are stock ratios, not averages of individual households’ price-per-room ratios.

What the reference rent $\bar{r}$ does. The supply equation uses a separate, fixed number $\bar{r} = 0.16$: $S(r) = H_0(r/\bar{r})^\xi$. Thus H_0 is the quantity supplied *at that reference rent*, not necessarily at the equilibrium rent r . On the same dollar conversion, $\bar{r}$ corresponds to \$184.12 per room per month; households need not pay that amount. It is not a price target or the earnings numeraire. It can be absorbed into the coefficient:

$$S(r) = Ar^\xi, \quad A = H_0/\bar{r}^\xi \simeq 19.966 \quad (H_0 = 6.294, \xi = 0.630).$$

This is the same supply curve. A third reference, r_* (code: `utility_reference_rent`), compensates family-dependent utility weights in the adopted reference; it is inactive in the floor specification. It is distinct from both actual rent r and supply reference rent $\bar{r}$.

Scope of the comparison. PSID earnings mainly concern the preceding tax year; AHS rents and self-reported values are not quality-adjusted. The floor point (no unsecured borrowing, chain 7, case 0173) remains experimental. Its preferences, entry/credit rules and calibration procedure differ; the price gap does not isolate one change. This preliminary check does not establish equivalence of the calibration procedures. The price-level readout and saved calculation retain full inputs and qualifications.